TukoSafe — Project Proposal

Moringa Hackathon 2026 · Human-Centered Technology: Building Trust in Communities (*People First, AI Second*)

Go out. Have fun. Get home safe. · Version 2.0 · 29 October 2026 · Supersedes Project Documentation v1.0

1. Project Overview

- **Project Name:** TukoSafe (from the Swahili *tuko salama*, "we are safe")
- **Team Name:** [Insert team name]
- **Challenge Area:** Community Safety & Wellbeing — youth social safety [confirm against the official challenge list]
- **Pilot Market:** Kenya, starting in Nairobi
- **Team Members & Roles:**
  - Lead / Product Planner: [Name]
  - User Researcher / Presenter: [Name]
  - Frontend / UI Designer: [Name]
  - Backend / AI Developer: [Name]

One-line pitch: TukoSafe is an AI-powered, people-led safety companion that helps friends in Kenya look out for each other on nights out: it notices when something may be wrong, helps a person ask for help discreetly, and gets everyone home.

Summary. TukoSafe combines voluntary, time-limited location sharing with scheduled check-ins, a discreet "Get Me Home" flow, SOS alerts and journey tracking. Version 1.0 of this project was a capable but rule-based safety app. Version 2.0 puts AI at the centre of the experience: an AI companion runs the check-ins in English, Swahili and Sheng, an AI copilot helps people ask for help without drawing attention, and an AI briefing turns a confusing missed check-in into a calm, actionable message for the friend who receives it. The principle is **AI does the work, people make the decisions.** AI never dispatches responders, never declares anyone "missing", and never tracks anyone who has not opted in. Every AI output can be ignored, edited, corrected or switched off, and the app still works when AI is unavailable.

TukoSafe is deliberately not a surveillance tool and does not replace professional emergency responders (999 / 112).

2. Problem Statement

Template: [Target users] experience [specific problem] because [reason or barrier]. This makes it difficult for them to [consequence or unmet need]. We want to explore a solution that helps them [desired improvement].

Project Statement:

Young adults aged 18–25 in Nairobi who go out at night in groups experience delays in noticing and responding to unsafe situations because friends get separated in crowded venues, phones die, and existing safety apps are built for family tracking rather than short, social, high-risk outings. This makes it difficult for them to ask for help discreetly, and for their friends to tell whether silence means an emergency or just a flat battery. We want to explore a solution that helps them look out for each other, request help without drawing attention, and get home safely, with AI reducing the effort and the false alarms while people stay in control.

Typical failure points

- Friends become separated in crowded venues.
- Someone leaves with a person nobody in the group knows.
- A friend becomes dangerously intoxicated or unwell.
- Someone goes missing or is involved in an accident on the way home.
- Nobody realises anything is wrong until hours later.

Why current tools fall short. Family-location apps such as Life360 offer opt-in sharing, check-ins and SOS, but they are built around family circles, not the short window of a night out. They also send raw, context-free alerts (“no location update”), which creates either panic or alert fatigue. Plain location tracking does not prevent harm. What is missing is *timely, understandable, trustworthy communication between people*, which is exactly where AI can help.

3. Target Users & Community Insights

Primary users: University students and young adults (18+) in Nairobi who go out in groups at night, to concerts, or on road trips.

Secondary users:

- **Safety contacts:** a friend, sibling or parent chosen to receive arrival confirmations and alerts.

- **Outing organisers:** the person who creates the group and agrees the plan.
- **Partner venue staff:** trained staff at clubs, restaurants and events (Plus phase).
- **Community moderators / admins:** review abuse reports, flagged AI outputs and venue applications.

Persona	Profile	Primary needs
Student on a night out	18–25, goes out in groups at weekends	Know friends are safe; leave a bad situation discreetly; get home safely
Safety contact	Friend, sibling or parent	Calm, clear alerts with last known location, time and caveats; an easy way to call or escalate
Outing organiser	Creates the group and plan	Fast setup and a clear view of who has checked in
Venue staff	Trained staff member	Simple guidance when a guest asks for help
Moderator / admin	TukoSafe support team	Abuse reports, audit logs, human review of flagged AI outputs

Community insights (hypotheses to validate on Thursday). We do not yet have primary research, so the following are assumptions we will test in 8–10 short interviews with peers (students, bootcamp learners, young professionals):

- Most people already text a friend “text me when you’re home”; the gap is what happens when the friend doesn’t reply.
- People avoid triggering alarms because they fear embarrassment or false alerts. Discreet, low-drama help matters more than loud alarms.
- Contacts who receive a vague alert at 2 AM don’t know whether to panic, so they either over-react or ignore it.
- Many users code-switch between English, Swahili and Sheng and expect the app to do the same.
- Users worry about being tracked by partners, family or strangers. Consent, visibility and auto-expiry are trust conditions, not extras.

Interview guide (five questions): (1) Tell me about the last time you went out at night. How did you get home? (2) Have you ever worried about a friend on the way home? What did you do? (3) What would make you actually use a safety app, and what would make you

delete it? (4) Who would you trust to see your location, and for how long? (5) If your friend's phone went silent at 2 AM, what would you want to be told?

Key user needs

- Discreet help that doesn't reveal anything publicly.
- Very low effort: one tap, or a few words in any language.
- Calm, specific alerts that say what is known, what is unknown and what to do next.
- Low data and low battery use, working on low-end Android devices and patchy networks.
- English, Swahili and Sheng-friendly language.
- Full control over who sees location, when it starts and when it ends.
- Clear honesty about limits: TukoSafe is not an emergency service.

4. Proposed Solution & Key Features

TukoSafe is an offline-tolerant, mobile-first web app (PWA) for the hackathon prototype, evolving into a native Android/iOS app for production. It lets a group create a time-boxed **Outing**, opt in to sharing for that window only, agree a **return-home plan**, and use an **AI companion** to check in, ask for help and keep friends informed.

Core Feature 1 (Demonstrable Prototype): Outing Circle + Msaidizi, the AI check-in companion. Friends create an outing, opt in to sharing for the outing window, and set a return-home plan. Msaidizi (Swahili for "helper") runs short, friendly check-ins in the user's language, adapts timing and grace period to the user's own habits (for example, how long this person usually takes to get home), and sends a gentle "Can you confirm you're okay?" instead of an alarm. Status chips are always paired with text: Green = checked in, Yellow = check-in overdue, Red = help requested. Copy states that status is not a guarantee.

Core Feature 2: Get Me Home Copilot. One prominent button opens three options: contact a trusted friend, share location and ask for help, or arrange a safe journey home. The AI copilot lets a user describe the situation in a few words, a code phrase or a tap ("I feel unsafe here", "Mambo si poa"), recommends the best of the three options, drafts a discreet message in the user's own voice for the friend to read, and suggests transport deep links. The user reviews and sends. One-tap 999 / 112 is always visible, with a notice that notifying friends is not dispatching responders.

Core Feature 3: AI Situation Brief for safety contacts + SOS. When a check-in is missed or SOS is pressed, the contact receives a plain-language brief generated from verified facts only: who, what happened, last known location and time, battery and connectivity caveats, the person's usual pattern, and recommended next steps (call first, message, check the map, and when to call 999 / 112). It always includes a caveat such as "This may be a dead battery or poor signal." SOS bypasses the escalation sequence and alerts immediately, with a press-and-hold countdown to prevent accidental triggers.

Core Feature 4: Trust Panel (transparency, oversight, feedback). Every AI output is labelled "AI-drafted", shows the facts it used ("based on: last location 1:42 AM, battery 8%, usual trip time 35 min"), and offers "Flag as wrong or unhelpful". Flags go to a human moderator queue. Users can switch each AI feature off independently.

Full feature set by phase

Area	Prototype (Friday)	MVP (v1)	Plus (later)
Accounts and trusted contacts	Anonymous demo session, simulated contacts	Phone OTP, invitations with consent, contact groups, instant revoke	Optional MFA, SMS links for contacts without the app
Outings and Safety Circles	Create outing, invite, opt in to sharing, auto-expiry	Full invitations, group map, leave anytime	Event partnerships
Check-ins and return-home plan	Msaidizi check-ins, grace period, escalation to contact	Configurable intervals, arrival confirmation	Personalised timing on device
Get Me Home	Copilot with three options (transport simulated)	Discreet messages, live location	Transport deep links, venue help
SOS	Press-and-hold, countdown, immediate alert, 999 / 112 dial	Offline queue, acknowledgement, silent mode	Verified response-provider dispatch
Journeys	Start journey, ETA prompt, "I've arrived"	Live journey monitoring	Informational route-deviation hints

Area	Prototype (Friday)	MVP (v1)	Plus (later)
AI Situation Brief	Generated from simulated facts	Live facts, SMS fallback	Multi-language refinement
Trust Panel	Labels, facts used, flagging, per-feature opt-out	Moderator console, audit log	Public transparency report
Safe Venue Network	Mock listing	Basic admin tooling	Verified venues, staff training, triage assistant

AI feature catalogue (what AI does and what people decide)

#	AI feature	What the AI does	What the human decides	Fallback if AI is unavailable
1	Msaidizi check-in companion	Runs conversational check-ins in English, Swahili and Sheng; suggests check-in timing	User confirms “I’m okay”, changes timing	Standard scheduled check-in with fixed templates
2	Get Me Home Copilot	Understands a short or coded request, recommends one of three options, drafts a discreet message	User edits and sends; chooses the option	Three static buttons with template messages
3	AI Situation Brief	Summarises verified facts for the contact, adds caveats and next steps	Contact decides whether to call, message or contact emergency services	Template alert with raw facts
4	Personal routine model (on device)	Learns the user’s typical trip duration and suggests a sensible grace period	User accepts or ignores; can reset	Default grace period

#	AI feature	What the AI does	What the human decides	Fallback if AI is unavailable
5	Journey anomaly hints	Notices a long stop or detour and asks the *user* “Everything okay?”	User responds; contacts are alerted only if the user asks for help or stays silent past the grace period	No hints; arrival confirmation only
6	Venue triage assistant (Plus)	Guides staff through the protocol for threatened, unwell or separated guests	Trained staff decide and act	Printed or in-app checklist
7	Coercion and abuse signals	Flags patterns such as repeated pressure to re-enable sharing	Human moderator reviews and acts	User reporting and manual review
8	Anonymous safety insights (Plus)	Aggregates anonymised patterns (for example, late-night overdue-check-in hotspots) with minimum group sizes	Organisers and admins decide on actions	Not shown

5. Role of Technology & AI (*People First, AI Second*)

Why this technology?

- **Progressive web app (React + Vite) for the prototype.** It installs from a link, runs on any low-end Android phone, needs no app-store approval and lets judges try it immediately. A service worker caches the app shell and queues check-ins and SOS requests while offline, which addresses the “patchy network” reality.
- **FastAPI backend.** Async endpoints, Pydantic validation (also used to validate AI output), automatic OpenAPI docs.
- **PostgreSQL + PostGIS (production).** Spatial queries for venue radius and last known location. The prototype uses SQLite and simulated locations to stay inside a one-day build.

- **WebSockets + Redis (production).** Live group status and alert acknowledgements; background workers (Celery or ARQ) for check-in timers, escalation and sharing expiry.
- **Push + SMS fallback.** Firebase Cloud Messaging with an approved SMS provider for critical alerts and for contacts without the app.
- **Native app later (React Native / Expo).** Reliable background location, foreground services during SOS and journeys, and push need a native-capable app. The PWA is the demonstrable first step, not the end state.

Role of AI: AI-powered, people-led

In TukoSafe, AI carries the *effort* (language, timing, summarising, drafting, noticing patterns) while people keep the *authority* (consent, decisions, escalation, accountability). The product would still work without AI, but it would be slower, noisier and less welcoming. That is the balance the theme asks for: AI is central to the experience and never central to the decision.

The AI ladder. Every AI feature stops at the lowest rung that is useful:

1. **AI informs:** summarises verified facts for a person.
2. **AI suggests:** recommends an option or drafts a message.
3. **Human approves:** the person edits, sends or ignores it.
4. **Human acts:** the person calls, messages or contacts emergency services.

Hard limits (never automated)

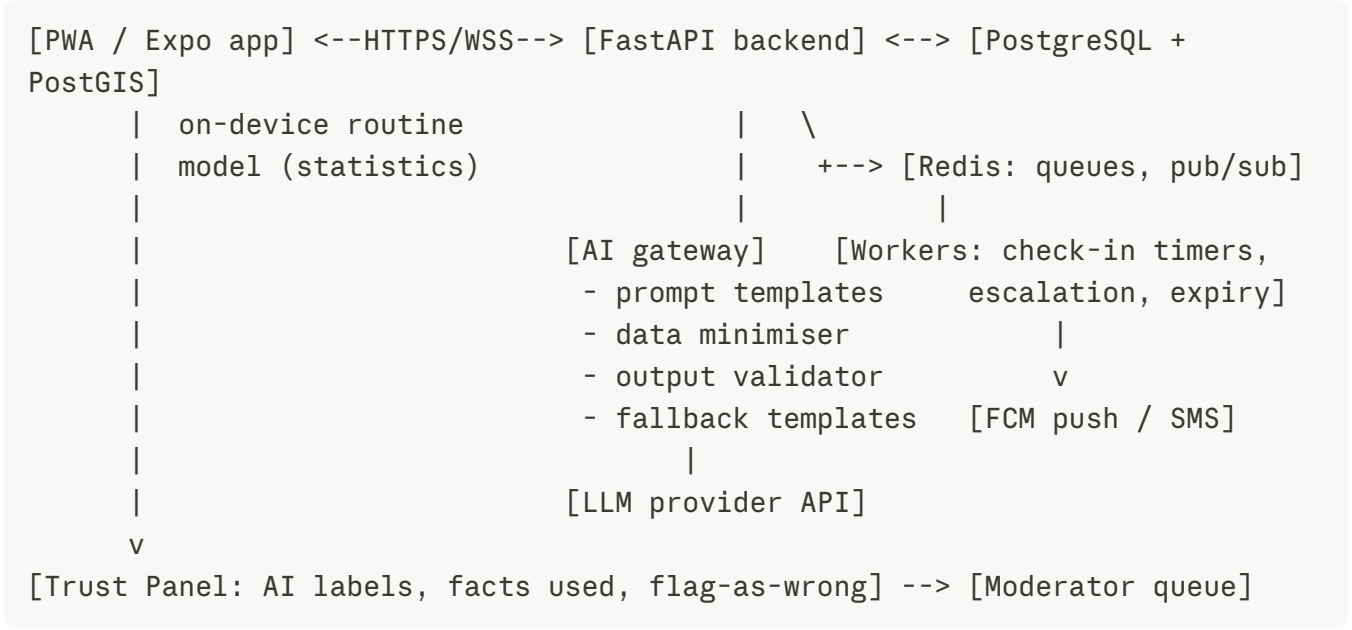
- AI never contacts police, ambulance or any responder, and never claims to.
- AI never labels a person “missing”, “in danger” or “unsafe”. It can only ask for verification.
- AI never starts or extends location sharing; only the user can, with explicit consent.
- AI never cancels or delays an escalation timer. Only the user’s own confirmation can.
- AI never sends a message the user has not seen, except the pre-agreed SOS and missed-check-in alerts whose templates the user approved in onboarding.
- AI never uses location history or conversations to train models or for advertising.

How each AI capability works

Capability	Inputs	Output	Guardrails
Msaidizi check-in companion	Language preference, return-home plan, time, current check-in state	Short check-in message in English, Swahili or Sheng; suggested grace period	Timers and escalation run in a deterministic scheduler. AI only words the message. Tone rules: no alarming language, always ask “Can you confirm you’re okay?”
Get Me Home Copilot	A few words of free text, a tap or a code phrase; coarse context (venue name, time)	Classification into one of three options plus a drafted discreet message	User text is treated as data, not instructions (prompt-injection safe). Raw coordinates are not sent to the model. The user edits and sends
AI Situation Brief	Structured JSON of verified facts: name, last location label, timestamp, battery, connectivity, usual trip time	Calm summary, caveats and next steps for the contact	Grounded generation. A validator checks that every time, place and number in the text matches the facts, and that the caveat and the 999 / 112 line are present. If the check fails, a template alert is sent instead
Personal routine model	The user’s past trip durations (on device)	Suggested grace period	Simple statistics (median and percentile), no cloud model, resettable at any time

Capability	Inputs	Output	Guardrails
Journey anomaly hints	Journey progress, stop duration, detour distance	Question to the user: “Everything okay?”	Rule-based thresholds decide when to ask; AI only phrases the question. Informational only
Coercion and abuse signals	Counts of repeated sharing requests, block and report events	Flag in the moderator queue	No automated penalties; a human reviews every flag
Venue triage assistant (Plus)	Request type, venue protocol	Step-by-step checklist in the staff member’s language	Trained staff decide; protocol text is authored and approved by humans

AI architecture



Model and tooling choices

- An LLM is accessed through a provider-agnostic AI gateway in the backend (for example, a Claude model through the Anthropic API). A small, fast model is chosen to keep responses under about two seconds.
- API keys live only in server environment variables or a secrets manager. They are never in the client, the repository or chat messages.

- The gateway enforces structured (JSON-schema) output, rate limits, a data minimiser that strips names, phone numbers and coordinates before any call, and fallback templates for when the model is slow, unavailable or fails validation.
- Prompt and response logging is off by default; only anonymised quality metrics (latency, validation pass/fail, fallback used) are stored.

Evaluation plan. We will build a test set of 40–60 scenarios in English, Swahili and Sheng and measure the following before the demo:

Measure	Target
Groundedness (briefs match the facts)	100% after the validator; failures fall back to templates
Tone (no alarming or blaming language)	Pass on review by at least three team members
Language quality (Swahili and Sheng)	Reviewed by native speakers; issues logged
Latency (95th percentile)	Under 2.5 seconds, otherwise fallback
Red-team checks	Prompt injection, coded or ambiguous requests and abusive use handled safely

6. Trust, Safety & Data Privacy

Transparency (provenance of every AI output and every venue)

The template asks that every resource shows where it came from. TukoSafe applies the same idea to everything a user is asked to trust:

- **AI outputs** carry an “AI-drafted” label, the facts they used (“based on: last location 1:42 AM, battery 8%”), a generated-at timestamp and a “Why am I seeing this?” explanation.
- **Locations** always show a last-updated time and the caveat that GPS can fail.
- **Venues** show a verification badge with verification date, reviewer, training status and next review date. A venue never receives certification for paying to be listed.
- **Status colours** (green, yellow, red) are always paired with text and icons, and the copy states they are not a guarantee.

Data minimisation

- **Prototype:** runs in the user's browser with a demo session. No registration, no personal tracking, no telemetry, no real location data.
- **Production:** phone number, display name and only the contacts the user adds. Location is collected only inside explicit, time-boxed sharing sessions that expire automatically, with short retention and a purge job.
- Granular consent for each sharing session that names exactly who sees what, and for how long. Periodic reminders while sharing is active.
- Only a minimised context goes to the AI provider (no coordinates, no phone numbers). No location histories are sold, ever, and no ads are targeted on movements.
- Users can export and delete their data and switch off each AI feature separately.

Human oversight & feedback

- Every AI output has "Flag as wrong or unhelpful". Flags go to a community moderator queue and feed the evaluation set.
- Humans make all consequential decisions: users approve their own messages; contacts decide how to respond; trained moderators review abuse flags; venue verification is done by people.
- Admin access to location data is role-based and written to an audit log.
- Alert wording is tested with real users for clarity and tone before launch.

Responsible AI commitments

- **Verify, don't assume:** a missed check-in triggers a request to verify safety, never an assumption of harm.
- **Honest about limits:** copy says plainly that TukoSafe is not an emergency service, notifying friends is not dispatching responders, and one-tap 999 / 112 is always available.
- **Fallback by design:** every AI feature has a non-AI fallback, so a model outage never blocks help.
- **Fairness across languages:** English, Swahili and Sheng are tested separately; known weaknesses are documented, not hidden.
- **Opt-out:** each AI feature can be switched off without losing core safety functions.

Abuse resistance (stalking and coercive control)

- No one can be added to a sharing group without accepting; stopping sharing never needs anyone else's permission.
- A visible "you are sharing with…" indicator at all times; block and report tools; rapid removal from any group.
- AI-assisted detection of coercive patterns (for example, repeated requests to re-enable sharing) routes to a human moderator and a support path. Support content explains controlling behaviour.

Security

- TLS everywhere, encryption at rest, secrets in a managed vault, short-lived JWTs with refresh rotation and device binding, optional MFA.
- Rate limiting on OTP, alert and AI endpoints; dependency scanning; an external security review before public launch.

Legal and regulatory context (Kenya)

TukoSafe handles sensitive personal data (precise location, contacts, possibly distress). Before public launch we plan to register with the Office of the Data Protection Commissioner under the Data Protection Act, 2019, define consent as the lawful basis, complete a Data Protection Impact Assessment, honour data-subject rights, set a minimum age of 18, and sign data-processing agreements with SMS, push and AI providers, including cross-border transfer checks. This section is a planning guide, not legal advice; a Kenyan data-protection lawyer should review it.

7. Friday Build Plan & Execution Roadmap

Thursday: Orientation & Discovery

- Refine the problem statement and confirm the challenge area with the organisers.
- Conduct 8–10 short interviews with peers using the guide in Section 3; capture quotes and note which hypotheses survive.
- Design wireframes for the five key screens: Home (Get Me Home + SOS), Outing Circle, Check-in conversation, Contact alert (AI Situation Brief) and Trust Panel.
- Set up the repository structure: `/frontend` (React + Vite PWA), `/backend` (FastAPI), `/ai` (prompts, validator, fallback templates, test scenarios), `/docs`.

- Set up the AI gateway with a test key stored in an environment file that is git-ignored; confirm the fallback templates work with no key at all.
- Write the first 40 test scenarios (English, Swahili, Sheng) and the Situation Brief fact schema.

Friday Morning (10:00 AM – 12:15 PM): Build

Time	Frontend / UI	Backend / AI	Presenter / Research
10:00–10:15	Kick-off; freeze scope to the cut line below	Kick-off	Kick-off
10:15–11:00	PWA shell, service worker, Outing Circle screen, status chips with text and icons	FastAPI endpoints for outings, return plans, check-ins, SOS; deterministic escalation timer	Draft the story arc and slides; collect interview quotes
11:00–11:45	Check-in conversation, Get Me Home screen (three options), SOS press-and-hold with countdown	AI gateway: Msaidizi prompts, Copilot classifier and drafts, JSON-schema output	Prepare the demo script and scenario data
11:45–12:15	Contact alert screen with AI Situation Brief, Trust Panel (AI label, facts used, flag, per-feature opt-out)	Situation Brief generator + validator + fallback templates; moderator flag endpoint	Draft the pitch deck

Friday Afternoon (12:15 PM – 1:30 PM): Test, Pitch, Submit

Time	Activity
12:15–12:40	Integrate and test local offline flows: queue a check-in and SOS while offline, confirm they sync; confirm the app works with the AI switched off
12:40–13:00	Run the scenario set; fix groundedness, tone and language failures; test a prompt-injection attempt

Time	Activity
13:00–13:15	Finalise the pitch deck and rehearse the 3-minute demo once
13:15–13:25	Finalise the README (setup, architecture, AI usage and limits, how to run the demo) and push to the repository
By 13:25	**Submit the repository link. The deadline is 1:30 PM.**

Scope cut line (what we drop first if time runs short)

1. **Must have:** Outing Circle, Msaidizi check-in with deterministic escalation, AI Situation Brief with validator and fallback, AI labels and “facts used”.
2. **Should have:** SOS with countdown and immediate alert, Get Me Home Copilot, offline queueing.
3. **Could have:** Flag-as-wrong and moderator queue, per-feature AI opt-out, journey “I’ve arrived”.
4. **Mock only:** transport deep links, venue listing, SMS delivery, map (simulated positions).

Demo script (about 3 minutes)

1. **Story (30 sec).** “Wanjiru” (an illustrative composite character, not a real person) goes to a concert with two friends. They agree an outing in TukoSafe and all opt in to sharing until 2 AM.
2. **Check-in (30 sec).** Msaidizi asks, in Sheng, if she is okay. She confirms.
3. **Discreet help (40 sec).** She feels uneasy at the venue, types a few words; the Copilot suggests “contact a trusted friend” and drafts a quiet message. She sends it.
4. **Missed check-in (40 sec).** Later her phone dies. The grace period passes, and her friend receives an AI Situation Brief: calm, factual, with last known location, “this may be a dead battery”, and next steps.
5. **Trust (30 sec).** We open the Trust Panel: AI label, facts used, flag-as-wrong, and the switch that turns AI off while core safety keeps working.
6. **Close (10 sec).** “AI does the work. People make the decisions.”

Pitch outline (mapped to the judging categories)

Slide	Content	Category it serves
1	The 2 AM problem: a silent phone and a friend who doesn't know what to do	Best Pitch & Storytelling
2	Problem statement and interview insights	Highest Community Impact
3	Live demo: check-in, discreet help, AI brief	Most Innovative Human-Centered Solution
4	People First, AI Second: the AI ladder, hard limits, fallbacks	Best Trust, Privacy & Responsible AI
5	Accessibility: Swahili/Sheng, low data, large targets, discreet mode	Best UX & Accessibility
6	Impact, business model, roadmap and the ask	Highest Community Impact; People's Choice

8. Potential Limitations & Future Scope

Current limitations (prototype)

- Locations, SMS delivery, transport links and the venue network are simulated; no real emergency-service integration exists, and TukoSafe is not an emergency service.
- A web app cannot guarantee background location on every phone. Reliable tracking, foreground services and push need the native app.
- AI features need connectivity. Offline, the app falls back to template messages and queues check-ins and SOS for sync.
- Sheng and code-switched Swahili are harder for language models; quality will vary and has not yet been validated with large groups of native speakers.
- AI generation adds cost and latency; a small model plus fallbacks keeps both bounded, but the cost model is unproven.
- Interview findings come from a small peer sample, and the legal review, DPIA and security review have not yet been done.
- GPS can be inaccurate, phones die and alerts can be missed. Status colours are not guarantees.

Future roadmap

Phase	Focus	Deliverables (AI highlighted)
0. Discovery (2–4 weeks)	Research and design	15–20 student interviews, legal review, consent and alert wording tests, clickable prototype, Swahili/Sheng evaluation set
1. MVP build (8–12 weeks)	Six core features	Auth, contacts, outings, temporary sharing, check-ins, SOS, journeys, admin basics, **Msaidizi, Copilot and Situation Brief in production**, AI gateway with validator
2. Closed pilot (6–8 weeks)	Nairobi volunteers	Measure alert delivery, battery and usability, **AI tone and false-escalation rate**, iterate on prompts and defaults
3. Plus features (8–12 weeks)	Safe Venue Network	Venue dashboard, verification workflow, transport links, separation alerts, **venue triage assistant, on-device routine model, anomaly hints, anonymous safety insights**
4. Partnerships	Scale	Universities, event organisers, transport partners, evaluation of emergency-response integrations, **fine-tuned or on-device language models for Sheng**

Durations are planning estimates and depend on team size.

Business model

Core safety features remain free; essential emergency functions stay accessible regardless of any paid tier.

Stream	Description
Venue partnerships	Verified listings, staff training and safety-management tools (verification is independent of payment)

Stream	Description
Event partnerships	Organisers sponsor safety features for concert and festival attendees
University partnerships	Institutions fund optional, consent-based student safety programmes
Corporate sponsorship	Brands fund public-safety initiatives without receiving user data
Premium convenience	Non-essential extras such as extended history and extra circles

Never: sell precise location histories or let advertisers target users by sensitive movements.

Success metrics (targets to validate in the pilot)

Metric	Target idea
Alert delivery success	Above 99% of SOS and escalation alerts delivered to at least one channel
Time to alert	SOS to contact notified in under 5 seconds on normal networks
Check-in completion	Majority of check-ins confirmed without escalation
False escalation rate	Tracked and reduced through better defaults and the personal routine model
AI quality	100% of briefs pass grounding validation; flagged-output rate falling over time
Adoption	Share of outings where all members join; repeat use per user
Trust	Users report feeling in control of their data; zero unresolved confirmed misuse incidents
Battery impact	Average drain during an outing within an agreed range

Risks and mitigations

Risk	Impact	Mitigation
Misuse for stalking or control	High	Mutual consent, visible sharing, blocking, auto-expiry, abuse reporting, human-reviewed coercion flags
False sense of security	High	Honest copy, no “safe” guarantees, 999 / 112 shortcuts, onboarding education
AI hallucination or alarming tone	High	Grounded generation, validator, fallback templates, tone tests, human flagging
Alert not delivered	High	Push plus SMS fallback, retries, monitoring, delivery audit
Background-location restrictions	High	Permission rationale, foreground services, clear tracking-unavailable states
Privacy breach or data sent to an AI provider	High	Data minimiser, no coordinates to the model, encryption, short retention, access audit, security review
Regulatory non-compliance	Medium	Data-protection registration, DPIA, legal review
Venue quality dilution	Medium	Independent verification, training and periodic review
Cost of maps, SMS and AI	Medium	Adaptive sampling, small models, cost modelling in the pilot
Low adoption (network effect)	Medium	Group-first onboarding, campus pilots, event partnerships

Open questions and next steps

- Minimum age and age-verification approach (18+ recommended).
- Android-first or simultaneous iOS release; map and SMS provider selection after cost comparison.
- Which AI model and hosting region to use, given data-protection and latency needs.

- Pilot partner: which university or venue to approach first.
- Whether any emergency-response provider will be approached for future dispatch.
- Next steps: validate with 15–20 interviews, test consent and alert wording, build the clickable prototype, engage a data-protection adviser and begin the DPIA, recruit pilot testers.

Appendix A. Requirements Compliance Matrix

Moringa Hackathon proposal template

Template requirement	Where it is covered	Change from v1.0
1. Project Overview (name, team, challenge area, roles)	Section 1	Added: team, challenge area, roles, pitch
2. Problem Statement (template and project statement)	Section 2	Added: template-format statement
3. Target Users & Community Insights, Key User Needs	Section 3	Added: insight hypotheses, interview guide, key needs
4. Proposed Solution & Key Features (demonstrable prototype)	Section 4	Added: prototype-scoped core features
5. Role of Technology & AI (People First, AI Second)	Section 5	New: v1.0 had no AI; AI ladder, hard limits, evaluation
6. Trust, Safety & Data Privacy (transparency, minimisation, human oversight)	Section 6	Added: AI provenance, flagging and moderation
7. Friday Build Plan & Execution Roadmap	Section 7	New: Thursday, Friday morning and afternoon plan, 1:30 PM deadline
8. Limitations & Future Scope	Section 8	Added: prototype limitations; v1.0 roadmap kept

Judging categories

Category	How TukoSafe addresses it
Most Innovative Human-Centered Solution	Event-based group safety with an AI companion, discreet help copilot and AI situation briefs for friends
Best Trust, Privacy & Responsible AI	AI ladder, hard limits, grounded generation with validator, fallbacks, data minimiser, flagging, opt-outs
Highest Community Impact	Addresses nightlife and travel risk for Kenyan youth; Safe Venue Network; university and event partnerships
Best UX & Accessibility	Swahili/Sheng, large touch targets, status colours with text, low-data mode, discreet mode, one-tap Get Me Home
Best Pitch & Storytelling	Story-led demo and slide map in Section 7
People's Choice	A relatable problem and a demo anyone can try on their phone

Appendix B. Detailed Functional Requirements

Priority: M = Must (MVP), S = Should (v1.x), C = Could (later). Requirements added in v2.0 for AI are marked **AI**.

ID	Requirement	Pri
FR-1.1	Register with phone number (OTP); optional email and multifactor protection	M
FR-1.2	Profile with display name and optional photo	M
FR-1.3	Add trusted contacts into Close Friends, Family and Emergency Contacts groups	M
FR-1.4	A contact must accept an invitation before receiving any alert or location	M
FR-1.5	Contacts without the app receive SMS alerts with a time-limited link to last known location	S

ID	Requirement	Pri
FR-1.6	Remove a contact or revoke access instantly	M
FR-2.1	Create an outing with venue, date and time window	M
FR-2.2	Invite by phone number or link; members see who joined, the venue and the return-home plan	M
FR-2.3	Per-outing opt-in to sharing; sharing ends automatically at the outing end time	M
FR-2.4	Status indicators: green, yellow, red, with copy that status is not a guarantee	M
FR-2.5	Group map shows only members who enabled sharing, with last-updated time	M
FR-2.6	Any member can leave and stop sharing at any time without giving a reason	M
FR-3.1	Record venue, expected departure, transport and the contact to notify on arrival	M
FR-3.2	Scheduled “Are You Okay?” check-ins with configurable interval and grace period	M
FR-3.3	Escalation: reminder to user, then notification to the safety contact after the grace period	M
FR-3.4	Contact sees last known location, timestamp and battery/connectivity caveats	M
FR-3.5	“I’m home” closes the plan and notifies the contact	M
FR-3.6	Timings are user-configurable, not hard-coded	M
FR-3.7 **AI**	Msaidizi delivers check-ins in the user’s language (English, Swahili, Sheng) with a non-AI template fallback	M
FR-3.8 **AI**	On-device routine model suggests a grace period from the user’s own trip history; user can accept, ignore or reset	S

ID	Requirement	Pri
FR-4.1	One prominent button opens three options: contact a friend, share location and ask for help, arrange a safe journey	M
FR-4.2	Option 1 sends a discreet message that does not reveal details publicly	M
FR-4.3	Option 2 alerts designated contacts with current location when available	M
FR-4.4	Option 3 shows transport options (deep links), lets the user share trip details and follow the journey	S
FR-4.5	States that notifying friends is not dispatching responders; one-tap 999 / 112	M
FR-4.6	Optional disguised UI (quiet haptics, no loud sounds)	S
FR-4.7 **AI**	Get Me Home Copilot interprets a short, coded or free-text request, recommends an option and drafts a discreet message for the user to edit and send	M
FR-5.1	Press-and-hold SOS with countdown and cancel	M
FR-5.2	SOS alerts designated contacts immediately, bypassing the missed-check-in sequence	M
FR-5.3	Shares live location and allows updates while SOS is active	M
FR-5.4	Silent mode; shortcut to call 999 or 112	M
FR-5.5	Recipients can acknowledge the alert	S
FR-5.6	Professional dispatch only through a separate, verified arrangement with a response provider	C
FR-5.7 **AI**	Every contact alert includes an AI Situation Brief grounded in verified facts, with validator and template fallback	M

ID	Requirement	Pri
FR-6.1	Start a journey with destination, start time, expected arrival, live location and monitoring contact	M
FR-6.2	If the journey runs late, prompt the user to check in; offer help options if they report a problem	M
FR-6.3	Route deviation hints, informational only	C
FR-6.4	Journey ends on arrival confirmation or manual end	M
FR-6.5 **AI**	Anomaly hints (long stop, detour) ask the user “Everything okay?” and never alert contacts on their own	C
FR-7.1	“I’ve lost the group” flag shares location with participating friends	S
FR-7.2	Group proposes and confirms a meeting point on the map	S
FR-7.3	The system never auto-classifies someone as missing because they moved away	M
FR-8.1 **AI**	Every AI output shows an “AI-drafted” label, the facts used and a way to flag it as wrong	M
FR-8.2 **AI**	Each AI feature can be switched off independently without losing core safety functions	M
FR-8.3 **AI**	Flagged outputs enter a moderator queue and the evaluation set	S
FR-8.4 **AI**	Coercion-pattern signals (for example, repeated requests to re-enable sharing) go to a human moderator, never to automatic penalties	S
FR-9.1	Push notifications with SMS fallback for critical alerts	M
FR-9.2	Admin dashboard: user support, abuse reports, venue applications, audit log viewer	M
FR-9.3	Partner venue dashboard: staff roster, incoming safety requests, protocol checklist (Plus)	C

ID	Requirement	Pri
FR-9.4 **AI**	Venue triage assistant guides staff through the protocol in their language (Plus)	C

Alert escalation and check-in state machine

Time or event	TukoSafe response
Expected arrival time passes	Msaidizi sends the user a check-in reminder
No response after the agreed grace period	The user's selected safety contact receives an AI Situation Brief
Contact receives the alert	Contact can call, message or view the last available location and timestamp
Contact has credible reasons to believe the user is in danger	Contact is prompted to contact the appropriate emergency services (999 / 112)
User activates SOS at any time	Urgent alert sent immediately, with no waiting for the missed-check-in sequence

```
SCHEDULED -> REMINDER_SENT -> GRACE_PERIOD -> CONTACT_ALERTED -> RESOLVED
   |              |               |                |
   +---- user confirms safe at any point ----------+

SOS_TRIGGERED -> CONTACT_ALERTED (immediate) -> ACKNOWLEDGED -> RESOLVED
```

A missing location update is not a confirmed emergency. Dead batteries, poor networks and disabled permissions can all explain silence, so alert wording asks for verification and never announces harm.

Appendix C. Non-Functional Requirements

Area	Requirement
Reliability	Target 99.5% monthly availability for alert services; queue-backed retries; push with SMS fallback

Area	Requirement
Performance	Location updates visible to the group in about 5 seconds on good networks; SOS dispatched within 3 seconds of confirmation; AI responses under about 2.5 seconds or fallback
AI safety	Structured output only; grounding validator on all contact-facing text; no AI step on the critical path of SOS delivery
Battery	Adaptive location frequency (higher during SOS or journeys, lower when idle); clear battery messaging
Connectivity	Offline queue for SOS and check-ins, last-known-location semantics, SMS fallback
Platform	Android first, iOS second; follow OS background-location rules; permission rationale screens
Security	TLS, encryption at rest, least privilege, audit logs, rate limiting, secrets management
Privacy	Automatic expiry of sharing sessions, configurable retention, export and delete tools
Accessibility	Large touch targets, high contrast, English and Swahili (Sheng-friendly tone), low-data mode
Scalability	Stateless API instances, Redis pub/sub for WebSockets, PostGIS spatial indexing

Appendix D. Data Model

All timestamps are UTC; sharing sessions carry an explicit `expires_at`.

Table	Key fields	Purpose
users	id, phone, display_name, email?, locale, mfa_enabled, ai_features (jsonb), created_at	Accounts and per-feature AI opt-in

Table	Key fields	Purpose
devices	id, user_id, fcm_token, platform, last_seen_at	Push targets
trusted_contacts	id, owner_id, contact_user_id?, contact_phone, group_type, status	Contacts and consent state
outings	id, creator_id, title, venue_id?, venue_name, starts_at, ends_at, status	Safety Circle events
outing_members	outing_id, user_id, role, joined_at, left_at	Membership
return_plans	id, outing_id, user_id, venue, depart_at, eta_home_at, transport, notify_contact_id	Return-home plans
location_shares	id, user_id, scope (outing/journey/sos), audience, started_at, expires_at, ended_at	Consent records for sharing
location_points	id, share_id, geom, accuracy_m, battery_pct?, recorded_at	Location updates (short retention)
checkins	id, user_id, plan_id, due_at, grace_until, responded_at, state	Check-in schedule and state
sos_events	id, user_id, triggered_at, cancelled_at?, resolved_at?, mode	SOS records
alert_deliveries	id, source_type, source_id, recipient_id, channel, sent_at, delivered_at?, ack_at?	Delivery and acknowledgement audit
journeys	id, user_id, destination, started_at, eta_at, ended_at, monitor_contact_id	Journey tracking
ai_outputs	id, feature, user_id, facts_hash, validator_result, used_fallback, latency_ms, created_at	AI quality metrics only; no message text stored by default

Table	Key fields	Purpose
ai_flags	id, ai_output_id, flagged_by, reason, status, reviewed_by?, reviewed_at?	Human review of flagged AI outputs
routine_stats (on device)	typical_trip_minutes, p90_trip_minutes	Personal routine model; never leaves the device
venues	id, name, geom, status, verified_at, reviewed_at, safety_features (jsonb)	Partner venues
venue_staff	id, venue_id, user_id, training_completed_at	Trained staff
safety_requests	id, user_id, venue_id, type, state, created_at	Requests handled by venue staff (Plus)
audit_logs	id, actor_id, action, entity, entity_id, ip, created_at	Security and access audit

Appendix E. API Design

REST over HTTPS with JWT bearer authentication and a WebSocket channel for live updates.

Method	Endpoint	Purpose
POST	/auth/otp/request, /auth/otp/verify	Phone sign-in
GET/POST/DELETE	/contacts	Manage trusted contacts and invitations
POST	/outings	Create outing
POST	/outings/{id}/invite, /join, /leave	Membership
PUT	/outings/{id}/return-plan	Set return-home plan
POST	/shares	Start a location share (scope, audience, duration)

Method	Endpoint	Purpose
POST	/shares/{id}/points	Send location update
POST	/shares/{id}/stop	End sharing immediately
POST	/checkins/{id}/confirm	Respond “I’m okay”
POST	/get-me-home	Trigger option 1, 2 or 3
POST	/sos, /sos/{id}/cancel, /sos/{id}/ack	SOS lifecycle
POST	/journeys, /journeys/{id}/arrive	Journey tracking
GET	/venues?lat=&lng=&radius=	Nearby partner venues
POST	/ai/checkin-message	Msaidizi message for a check-in (returns text, language, fallback flag)
POST	/ai/get-me-home	Copilot recommendation and draft message (user reviews before sending)
POST	/ai/situation-brief	Grounded brief for a contact alert (validator result included)
POST	/ai/flag	Flag an AI output as wrong or unhelpful
PUT	/settings/ai	Switch individual AI features on or off
WS	/ws/outings/{id}	Live group status and locations

Appendix F. Safe Venue Network (Plus)

A TukoSafe Partner Venue agrees to a defined safety protocol, linking digital tools to real-world help. It is expected to be the strongest long-term differentiator.

Venue requirements: a designated, trained staff member to receive safety requests; a private place for a guest to wait for assistance; procedures for helping a guest who feels threatened or unwell; clear steps for contacting emergency services; procedures for helping a guest reconnect with trusted friends.

Stage	Description
Apply	Venue submits application and documentation
Verify	TukoSafe checks identity, licensing and protocol readiness
Train	Staff complete safety training; completion is recorded
Listed	Venue appears in the app with its available safety facilities and verification badge
Review	Periodic review, user feedback and spot checks; listing can be suspended

A venue must never receive safety certification simply for paying to be listed. Payment may cover training and tools, but verification and review are independent of payment. The AI triage assistant supports staff but never replaces their training or judgement.

Appendix G. Sample AI Message Wording and Glossary

Tone is calm and verification-first. These are examples of what the AI is guided to produce; the validator blocks anything that adds facts not in the data.

Situation	Example message
Msaidizi check-in (English)	Hi Brian, are you home safe? Please tap to confirm so your friends know you’re okay.
Msaidizi check-in (Swahili)	Habari Brian, umefika nyumbani salama? Tafadhali thibitisha ili marafiki zako wajue uko sawa.
Discreet help request (drafted by Copilot)	Your friend has asked you to call or come and help. Tap to see where they are.

Situation	Example message
AI Situation Brief (missed check-in)	Brian hasn't confirmed arrival yet. This may be a dead battery or poor signal. Last known location: Westlands, 1:42 AM; battery was 8%. His trips home usually take about 35 minutes. Please try calling him first. If you believe he is in danger, call 999 or 112. *AI-drafted from the facts shown; tap to flag.*
SOS	Brian has requested urgent help. Last location shared at 1:42 AM. Call him now. If you believe he is in danger, call 999 or 112.

TukoSafe is not an emergency service. In an emergency, call 999 or 112.

Term	Meaning
Safety Circle	Group of people in an outing who have opted in to share status and location
Return-home plan	Planned departure, transport, ETA and contact to notify on arrival
Grace period	Time allowed after a check-in is due before the contact is alerted
Get Me Home	Discreet assistance flow with three help options, supported by the AI Copilot
Msaidizi	The AI check-in companion (Swahili for "helper")
AI Situation Brief	A grounded, plain-language summary sent to a safety contact
AI ladder	Inform, suggest, human approves, human acts: AI never goes past suggesting
Partner Venue	Verified venue with trained staff and a defined safety protocol